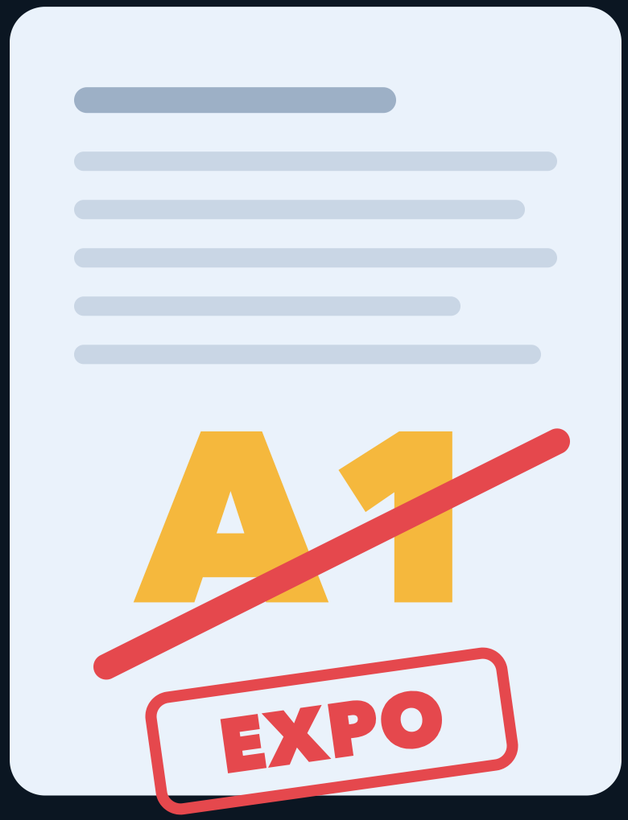
WEEK 5 · DAY 1

No Expo! Honest Splits + Quiz 3

If you saw the questions before the exam,
the A1 is fake. Today our detectors sit a real exam. 📝



NIGCOMSAT
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Publica Academy

TODAY'S MISSION · 2 HOURS

Spot the expo. Then a fair fight.



Spot the expo

why random splits lie



Split by time

train · embargo · test



Fair fight

forest vs Week 4 chart



Quiz 3

the machine-learning days

⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 45 min hands-on · 📝 20 min quiz · 🏁 15 min mark + wrap

QUICK QUESTION

Who do you put in the control room? 🛰️

🤩

Tunde: A1 in every subject

The night before WAEC, Tunde's cousin brought **"expo"**: the actual questions. Tunde memorised the answers.

perfect score | saw the questions

📚

Chika: B2 in maths

Chika studied past questions, then sat an exam she had **never seen**.

honest score | a bit lower

Tunde's A1 measured the expo, not Tunde.
A detector tested on hours it has already seen is **Tunde**. 🙅🏾

EXPO IN TIME SERIES COMES IN THREE FLAVOURS

Three ways a score gets faked



Neighbours in training

Shuffle the hours: the test hour at 10:00 has 09:00 and 11:00 in training. Almost the same reading.



Features that peek ahead

`shift(-1)` or `rolling(..., center=True)` use readings that don't exist yet.



Settings tuned on the key

Picking the threshold or contamination by looking at the **test** answers.

All three give a score that won't survive the **next real day** in the control room.

PART 1

Catch the expo 

One model, two ways of splitting the year.

A SIDE EXPERIMENT

Meet the copycat 🐱

For each new hour it finds the **one most similar hour** it has seen and copies that hour's answer: fault or normal.

```
copycat = make_pipeline(StandardScaler(),
            KNeighborsClassifier(n_neighbors=1))
copycat.fit(X_train, y_train)   # y = 1 inside a fault
copycat.predict(X_test)
```

It learns from the answer key (**supervised**). Our forest never sees labels. The copycat is only here to expose expo.



SAME MODEL · SAME FEATURES · SAME ANSWER KEY

How many fault hours did the copycat find?



Random split

100

% of 241 test fault hours · "A1" 🎉



Split by time

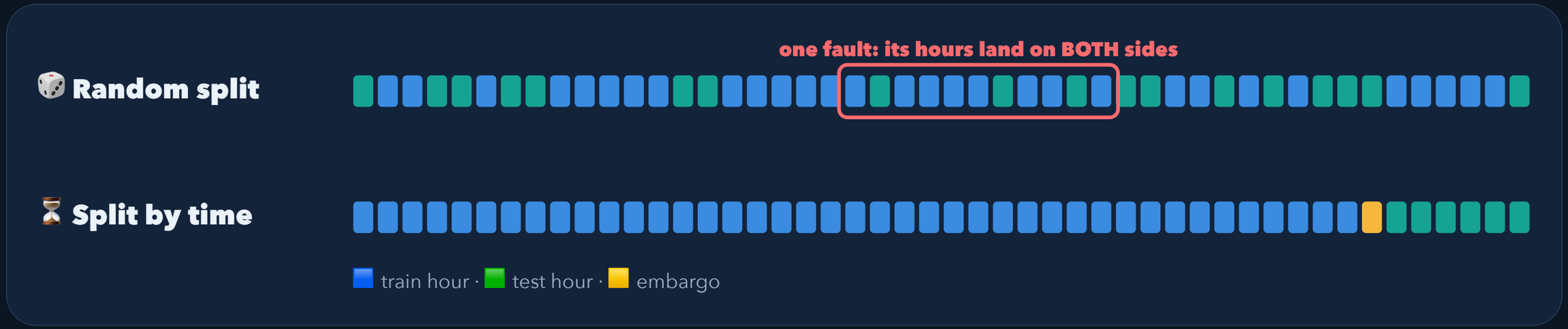
3.4

% of 696 test fault hours (24 found)

By time, it only caught **EV05**: a repeat of EV02, which it saw in May.
The only difference between 100 and 3.4 is **expo**.

WHY THE RANDOM SPLIT LIES

The test hours have twins in training 👥



EV04 lasts 672 hours. Shuffle them and every test hour inside it has its neighbour in training.
The copycat isn't detecting the fault, it's **copying the answer from the next seat**.

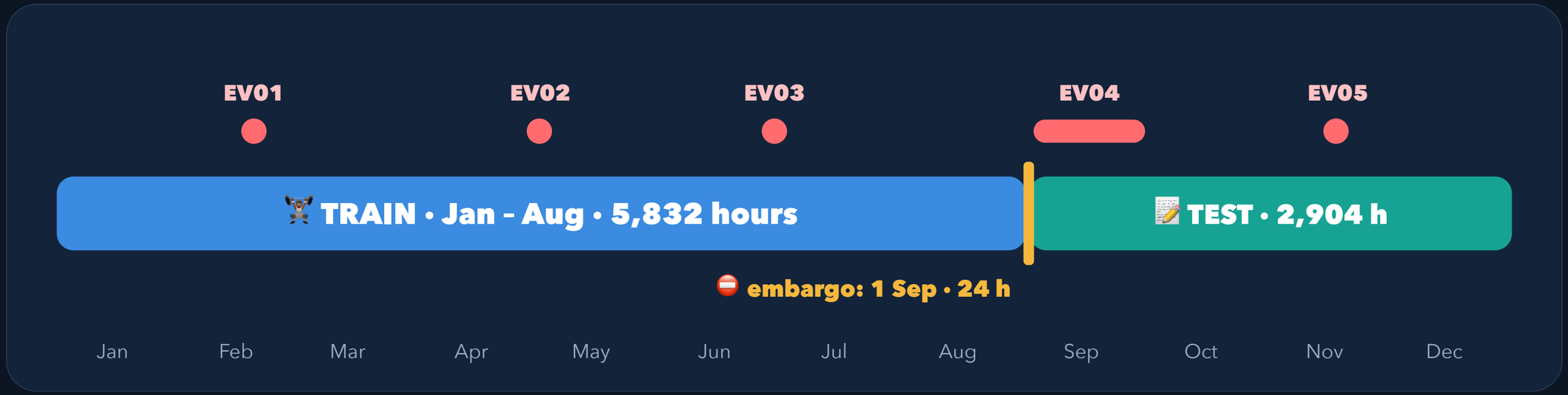
PART 2

Split by time ✂️

Train on the past. Test on the future. Leave a gap.

THE HONEST SPLIT FOR THE WHOLE WEEK

Past to train, future to test




```
train = bus.index <  "2025-09-01"     # Jan – Aug: 5,832 hours, EV01–EV03
test  = bus.index >= "2025-09-02"     # 2 Sep – 31 Dec: 2,904 hours, EV04 + EV05
assert bus.index[train].max() < bus.index[test].min()   # fails loudly if they ever overlap
```

WHY SKIP 1 SEPTEMBER?

The embargo: an empty seat between candidates 🪑



Without a gap, a 24-hour feature on the first test hour **reads training hours**. The test isn't separate any more.

Rule: embargo ≥ your longest feature window. Our longest is 24 hours, so we skip one day.

WRITE THE RULES BEFORE THE FIGHT

Four rules of a fair fight 👩🏾‍⚖️

1 Same held-out months

Everyone is judged on 2 Sep – 31 Dec only.

2 Same hours

Steady sunlight: eclipse and recharge have their own normal.

3 Same scorer

`score_events`: one flag inside = caught, outside = false alarm.

4 Settings fixed first

Chart k = 3 (Week 3). Forest contamination = 0.5% (Week 4 Day 2).

PART 3

The fair fight 🥊

Four contenders. Two events in the test months: EV04 and EV05.

CONTENDER 1 · THE WEEK 4 CHART EXACTLY AS TAUGHT (SUNLIT HOURS)

Caught 2/2… but look when 🍀



It lit **84 of 672** EV04 hours, and **every one** was 22:00, 23:00 or 00:00: the battery recharging after the 21:00 eclipse.

Plus **50 false alarms**: all of them recharge hours in the autumn eclipse season.

Right answer, wrong reason = a **lucky catch**. The amplifier fault had nothing to do with it.

EVERY ALARM IN THE TEST MONTHS · 2 SEP – 31 DEC

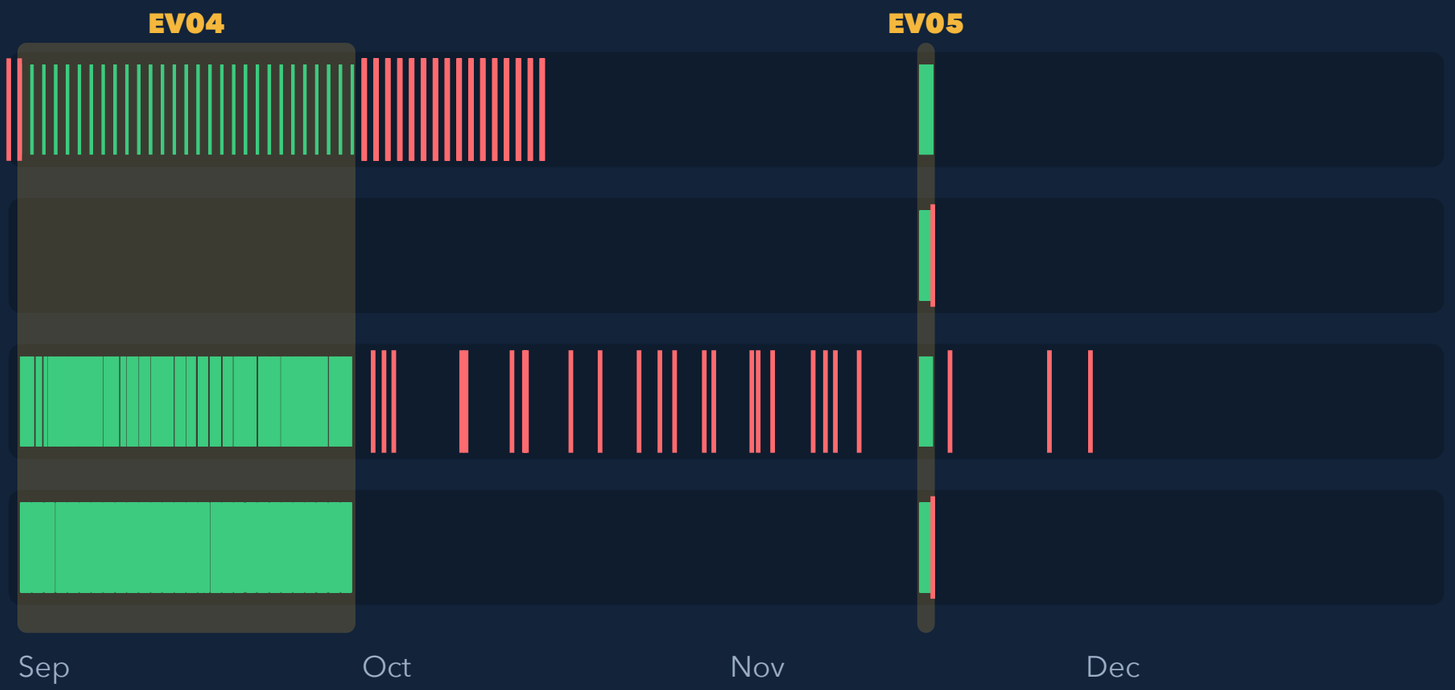
Same months, same hours, four detectors

1. Week 4 chart · raw · sunlit (as taught)
caught 2/2 · 50 false alarms

2. Week 4 chart · raw · steady sun
caught 1/2 · 1 false alarm

3. Isolation forest · lean features
caught 2/2 · 25 false alarms

4. Week 4 chart · lean features
caught 2/2 · 1 false alarm



tick = a flag inside a real event · tick = a false alarm · gold band = the event

THE REPORT CARD · TEST MONTHS ONLY

One scorer. Four contenders.

DETECTOR	CAUGHT	FALSE ALARMS	EV04 HOURS LIT	PRECISION
1. Week 4 chart · raw · sunlit (as taught)	2/2	50	84 / 672 🍀 lucky	0.04
2. Week 4 chart · raw · steady sun	1/2	1	0 / 672	0.50
3. Isolation forest · lean features	2/2	25	358 / 672	0.07
4. Week 4 chart · lean features	2/2	1	502 / 672	0.67

Lean features = the 4 health channels + **temp_per_drive** (Week 4 Day 3).

THE VERDICT

The feature won, not the forest 🏆

1

false alarm (forest: 25)

502

EV04 hours lit (forest: 358)

k = 3

the only setting, chosen in Week 3

We recommend the **simpler detector**. Saying “my fancy model lost” is a professional result, not a failure.

"BUT YOU CAN TUNE THE FOREST!"

Tune it honestly: still no win

FOREST BUDGET	PAST EVENTS (JAN-AUG)	TEST CAUGHT	TEST FALSE ALARMS	EV04 HOURS LIT
🔒 **Fixed before the test (Week 4 Day 2)** contamination = 0.50%	3/3	2/2	25	358 / 672
🤦🏾‍♀️ **Honest tune: past labels only** contamination = 0.35%	3/3	2/2	2	186 / 672
🙈 **Expo tune: picked by peeking at Sep-Dec** contamination = 0.30%	2/3 (misses EV03)	2/2	0	144 / 672
📊 **Week 4 chart, k = 3** no tuning at all	3/3	2/2	1	502 / 672

The 0-false-alarm budget exists only if you **peek at the test**, and it misses EV03 in the past.
At its honest best the forest still trails a chart that needed no tuning.

IS 25 JUST BAD LUCK?

Five seeds, five forests 🌲🌲🌲🌲🌲

random_state = 42	random_state = 0	random_state = 1	random_state = 7	random_state = 123
25	**25**	**36**	**20**	**28**
false alarms	false alarms	false alarms	false alarms	false alarms

From 20 to 36. The chart has no dice to roll: **1**, every time.

ONE CUT = ONE POSSIBLY LUCKY NUMBER

Rolling origin: four exams, not one 🔁

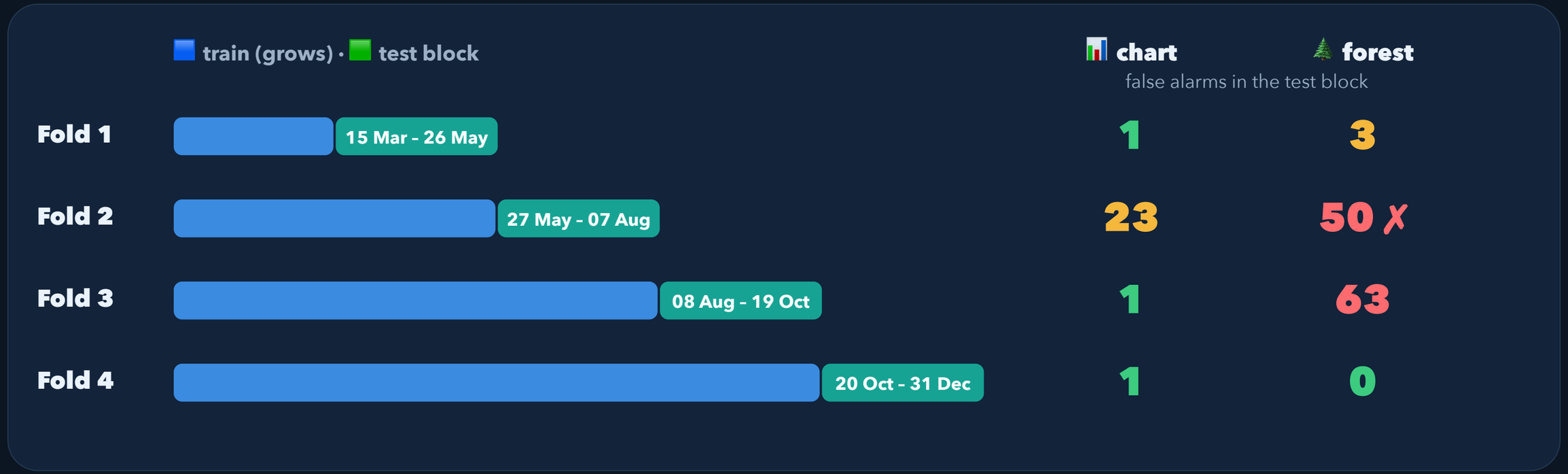
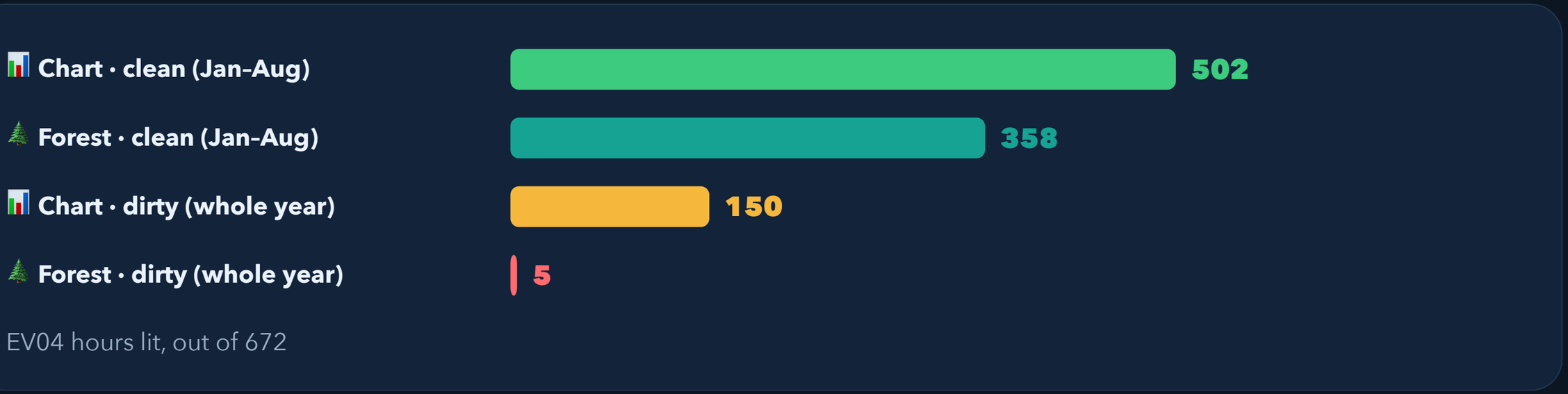


Chart: 1, 23, 1, 1 · forest: 3, 50, 63, 0 (✗ = missed the fold's event). Training only ever grows **forwards**. You get a spread, not one score.

WHERE DOES "NORMAL" COME FROM?

A dirty reference period makes the fault normal



Fit on a year that **contains** EV04 and the forest learns the hot amplifier as normal.
In production there are no labels: ask operations for a period they **believe was quiet**, and write it in your report.

SO IS MACHINE LEARNING USELESS? NO.

When does the forest earn its place?



Chart country (today)

6 channels. Known physics. One engineer can write the right feature and read every alarm.



Forest country

Hundreds of channels. Relationships nobody has written down. Faults that nudge many channels a little at once.

Either way the method is the same: **split by time, fix the settings, same scorer, honest verdict**.

QUIZ 3 · THE MACHINE-LEARNING DAYS

30 questions. 20 minutes. Just you. 

Phones down

notebooks closed, no AI

One letter

circle A, B, C or D (or True/False)

No “always B”

answers are spread across A–D

Mark together

swap papers and tick

Covers scikit-learn + forests + clustering (W4D2) · features (W4D3) · honest splits (today).

YOUR TURN · 45 MINUTES

Referee the fight 🥊

1. Open **W5D1-exercise**, save a copy, run STEP 0
- **A** Catch the copycat's expo
- **B** Build the split + embargo
- **C** Run the fair fight, pick a winner
- **D** Rolling origin + a dirty reference



14 checks to pass

🌶️ Pepper task: tune the forest the honest way, using only the past

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know



Expo is everywhere: random splits, features that peek, settings tuned on the test.



Split by time with an embargo ≥ your longest window, and `assert` it.



Fair fight, honest verdict: if the simple model wins or ties, recommend it.

Next 👉 **Forecasting baselines**: Grandma's "tomorrow will be like today". 👵🏾

Final project: bring your own split-by-time + baseline table to Checkpoint 4 (Week 5 · Day 3).